

Capstone Project Proposal

Forecasting Electric Vehicle (EV) Adoption and Charging Infrastructure Needs in Washington State (2025–2050)

As of December 2024, the U.S. Department of Energy reported 3.55 million registered electric vehicles (EVs) nationwide, with California, Washington, Florida, and Texas leading adoption. In Washington State, approximately 199,000 battery electric vehicles (BEVs) were registered by mid-2025 among 3.77 million residents aged 25–59, representing the most active driving population. Washington’s Clean Energy Transformation Act targets 45% EV adoption by 2030, 70% by 2040, and 95% by 2050, along with 80% residential home charging access by 2030. However, charging infrastructure expansion has lagged behind these ambitious goals, especially in suburban and rural counties. This project addresses: **How can Washington State optimize its public and private charging network to sustain EV adoption growth through 2050?**

2. Data Sources and Acquisition

- This project integrates multiple datasets:
- EV Registration (199,000 VIN-level BEV records) — vehicle make/model, geocoded coordinates by county.
 - Charging Network — Tesla Supercharger and public DC fast-charging data from Tesla Map & AFDC API.
 - Population & Housing Data — county-level demographics (ages 25–59) and household counts from WA OFM and U.S. Census.
 - Policy & Incentives — Clean Energy Act milestones and EV rebates from WA Department of Commerce.

All datasets are publicly available or downloaded in Excel/CSV format for modeling and visualization.

3. Analytical Approach

This is a **supervised regression problem** predicting EV adoption (target) based on charger density, demographics, and policy variables. **Predictors:** population (25–59), charger density (public + home), urban/rural type, income levels, corridor proximity, and incentives.

Framework: spatial analysis, regression, and forecasting with scenarios:

- Baseline — current charger expansion rate.
- Policy-Aligned — accelerated build-out to reach 95% adoption by 2050.
- Home-Charger Heavy — 80% household charger saturation reduces public need.

Models: Linear Regression, Random Forest, XGBoost, and logistic growth forecasting.

4. Final Deliverables

Final outputs will include:

- Interactive map of EVs and chargers by county.
- Predictive model linking charger build-out to EV adoption milestones (2030–2050).
- Scenario visualizations and investment recommendations.

Deliverables will be shared as a Jupyter Notebook, Power BI dashboard, and optional Flask web app.

5. Computational Resources

Resource Type	Estimated Need	Purpose
CPU	4–8 cores	Spatial joins, regression training
Memory (RAM)	16–32 GB	Handle 200k+ VIN records and mapping
GPU (Optional)	1 × NVIDIA T4	Accelerate model training

Storage	10–20 GB	Datasets, outputs, and map exports
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Summary

This capstone will model Washington’s trajectory to **95% EV adoption by 2050**, incorporating demographic, infrastructure, and policy data. By simulating charger deployment scenarios—including 80% home-charging penetration by 2030—this project aims to deliver actionable, data-driven recommendations for state policy, utilities, and private charging providers, supporting Washington’s goal of a fully electric, carbon-neutral transportation system.